The empirical recurrence for OEIS A283638: recovering a conjecture that is not written down

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Abstract

OEIS A283638 records "Empirical recurrence of order 63", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 176$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A283638 is "Number of 5Xn 0..1 arrays with no 1 equal to more than one of its horizontal, diagonal and antidiagonal neighbors, with the exception of exactly two elements.". It has offset 1 and begins

$$0, 142, 3729, 85469, 1185010, 19517898, 272974255, 3604990867, \ \ldots$$

The entry states:

> Empirical recurrence of order 63 (see link above)

As of the "Last modified" line on the live entry (Mar 13 2017 04:28:15, revision 7) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1559$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 176$ of the 1559, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 176.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 352$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{63} c_i \, a(n-i),$$

c_1	12	c_{17}	143718459201	c_{33}	−1255717070235329	c_{49}	−17711232882264
c_2	78	c_{18}	−472111101157	c_{34}	1181425845979569	c_{50}	138482024915250
c_3	−350	c_{19}	1488623093958	c_{35}	−683096365449762	c_{51}	−71287526892024
c_4	−8220	c_{20}	−4282648609737	c_{36}	−401610502187513	c_{52}	−7686911763000
c_5	−14913	c_{21}	11848698629857	c_{37}	1386102281112321	c_{53}	23958978676773
c_6	141351	c_{22}	−29694937325805	c_{38}	−1365399367213497	c_{54}	−32133090640501
c_7	1408809	c_{23}	70782497452332	c_{39}	1040997528437205	c_{55}	−1756476169746
c_8	4229631	c_{24}	−158442277627613	c_{40}	−76470590446515	c_{56}	13789803375
c_9	13806030	c_{25}	325665045618522	c_{41}	−547794045717843	c_{57}	94928075703
c_{10}	18635244	c_{26}	−608160396199332	c_{42}	696426993040135	c_{58}	3213114549
c_{11}	105368541	c_{27}	982303867814798	c_{43}	−718573353371583	c_{59}	−119946834
c_{12}	−148146550	c_{28}	−1340292587984178	c_{44}	−14962955305011	c_{60}	−89593471
c_{13}	627924012	c_{29}	1467865212826329	c_{45}	276247285232107	c_{61}	−1430910
c_{14}	−3592155210	c_{30}	−1125399297285032	c_{46}	−384439245790110	c_{62}	78300
c_{15}	10888496611	c_{31}	367774846954065	c_{47}	288274577465532	c_{63}	27000
c_{16}	−46869552186	c_{32}	591826595413341	c_{48}	−789189682712		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63 + 4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

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